Austo _Motor_Company_Report



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- 1. Do men tend to prefer SUVs more compared to women?
- 2. What is the likelihood of a salaried person buying a Sedan?
- 3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?
- 4. How does the the amount spent on purchasing automobiles vary by gender?
- 5. How much money was spent on purchasing automobiles by individuals who took a personal loan?
- 6. How does having a working partner influence the purchase of higher-priced cars?

1 Introduction and about the Data

Introduction

Austo Motor Company is a car Manufacturer, Manufacturing 3 types of vehicles, namely `SUV` , `Sedan` , and `Hatchbasck`

In a Board meeting a concern regarding the efficiency of the Marketing campaign being used along with getting an Ideal about the demand of the customer in order to Improve Customer Experience

So therefore they want to improve :

- Marketing Campaign strategy
- Customer Experience

Data Description and Classification

- **Age**: The age of the individual in years.
- **Gender**: The gender of the individual, categorized as male or female.
- **Profession**: The occupation or profession of the individual.
- **Marital_status**: The marital status of the individual, such as married &, single
- **Education**: The educational qualification of the individual Graduate and Post Graduate
- **No_of_Dependents**: The number of dependents (e.g., children, elderly parents) that the individual supports financially.
- **Personal_loan**: A binary variable indicating whether the individual has taken a personal loan "Yes" or "No"
- **House_loan**: A binary variable indicating whether the individual has taken a housing loan "Yes" or "No"

- **Partner_working**: A binary variable indicating whether the individual's partner is employed "Yes" or "No"
- **Salary**: The individual's salary or income.
- **Partner_salary**: The salary or income of the individual's partner, if applicable.
- **Total_salary**: The total combined salary of the individual and their partner (if applicable).
- **Price**: The price of a product or service.
- **Make**: The type of automobile

Give above after having gone through the description of each Variable, given that our Objective is it Improve Marketing Strategy and Customer Experience, our Target Variables will be `Price`, and `Make`

Regarding the rest of the Variables they are Independent Variables and all of them are Significant as their relation to `Price` and `Make` are meaningful

Lets Identify the Type of each Variable

Variable	Type
Age	Numerical
Gender	Categorical
Profession	Categorical
Marital_status	Categorical
Education	Categorical
No_of_Dependents	Numerical
Personal_loan	Categorical
House_loan	Categorical
Partner_working	Categorical
Salary	Numerical
Partner_salary	Numerical
Total_salary	Numerical
Price	Numerical
Make	Categorical

2 Data Overview, Processing, and Cleaning

Structure of Data

Here we have 1581 records with 14 columns

Checking Data Types

```
Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Age               1581 non-null   int64  
 1   Gender            1528 non-null   str    
 2   Profession        1581 non-null   str    
 3   Marital_status    1581 non-null   str    
 4   Education         1581 non-null   str    
 5   No_of_Dependents  1581 non-null   int64  
 6   Personal_loan     1581 non-null   str    
 7   House_loan        1581 non-null   str    
 8   Partner_working   1581 non-null   str    
 9   Salary            1581 non-null   int64  
 10  Partner_salary    1475 non-null   float64
 11  Total_salary      1581 non-null   int64  
 12  Price             1581 non-null   int64  
 13  Make              1581 non-null   str    
```

Fig1

Checking for and Treating Missing values and Data irregularities

Here we checked the Number of Unique values in each Column

```
Age                 33
Gender               4
Profession           2
Marital_status       2
Education            2
No_of_Dependents     5
Personal_loan        2
House_loan           2
Partner_working      2
Salary             538
Partner_salary     149
Total_salary       754
Price               53
Make                 3
dtype: int64
```

Fig2

First thing we Notice is Gender having 4 Unique values where its supposed to be 2 for `Male` and `Female` . Lets Investigate it.

```
<StringArray>
['Male', 'Femal', 'Female', nan, 'Femle']
Length: 5, dtype: str
```

Fig3

After use of the replace method to replace the incorrect spellings with the correct ones we now have 3 distinct values which include Nan as shown below

```
['Male', 'Female', nan]
Length: 3, dtype: str
```

Fig4

Here it was decided to replace the Nan missing values of Gender with the modal values ie most occuring value and given that there are only 53 missing values out of 1581 it makes sense to replace with most occuring value which would be `Male`

```
Age                   0
Gender               53
Profession            0
Marital_status        0
Education             0
No_of_Dependents      0
Personal_loan         0
House_loan            0
Partner_working       0
Salary                0
Partner_salary      106
Total_salary          0
Price                 0
Make                  0
dtype: int64
```

Fig5

We will now Discuss how to deal with missing values of `Partner_salary` , as we can see we have a variable by the name of `Partner_working` which we can use to fill out the Nan values of `Partner_salary` according to the following conditions for the 2 cases

Filling Partner_salary According to Parnter_working Status of 'Yes' or 'No'

- If '`Yes`' Partner_salary = Total_salary - Salary
- If '`No`' Partner_salary= 0

But before that we find out how many such values for each case exists

```
16 rows in Partner_salary with Parnter working as 'Yes'
90 rows in Partner_salary with Parnter working as 'No'
```

Fig6

Now we fill up the Nan missing values of `Partner_salary` According to the above mentioned conditions

```
Age                  0
Gender               0
Profession           0
Marital_status       0
Education            0
No_of_Dependents     0
Personal_loan        0
House_loan           0
Partner_working      0
Salary               0
Partner_salary       0
Total_salary         0
Price                0
Make                 0
dtype: int64
```

Fig7

We now no longer have any Missing values

Also the Number of Duplicate Entries are 0

Observations and Insights

Let us now see the Statistical Summary of all the Numerical variables

	count	mean	std	min	25%	50%	75%	max
Age	1581.0	31.922201	8.425978	22.0	25.0	29.0	38.0	54.0
No_of_Dependents	1581.0	2.457938	0.943483	0.0	2.0	2.0	3.0	4.0
Salary	1581.0	60392.220114	14674.825044	30000.0	51900.0	59500.0	71800.0	99300.0
Partner_salary	1581.0	19233.776091	19670.391171	0.0	0.0	25100.0	38100.0	80500.0
Total_salary	1581.0	79625.996205	25545.857768	30000.0	60500.0	78000.0	95900.0	171000.0
Price	1581.0	35597.722960	13633.636545	18000.0	25000.0	31000.0	47000.0	70000.0

Fig8

- Average age of the Customer is around 32 , youngest being 22 and eldest being 54
- The Number of Dependents seem to average around 2-3 people , possibly indicating new couples about to start or have started a new family
- Average Salary is around 60,000, while Partner Salary averages around 20,000 and Total Salary is averaging around 80,000, but Partner Salary seems to be showing significant amount of variability.
- Average Price for vehicles is around 35,000-36,000

3 Exploratory Data Analysis

3.1Univariate Analysis

Age

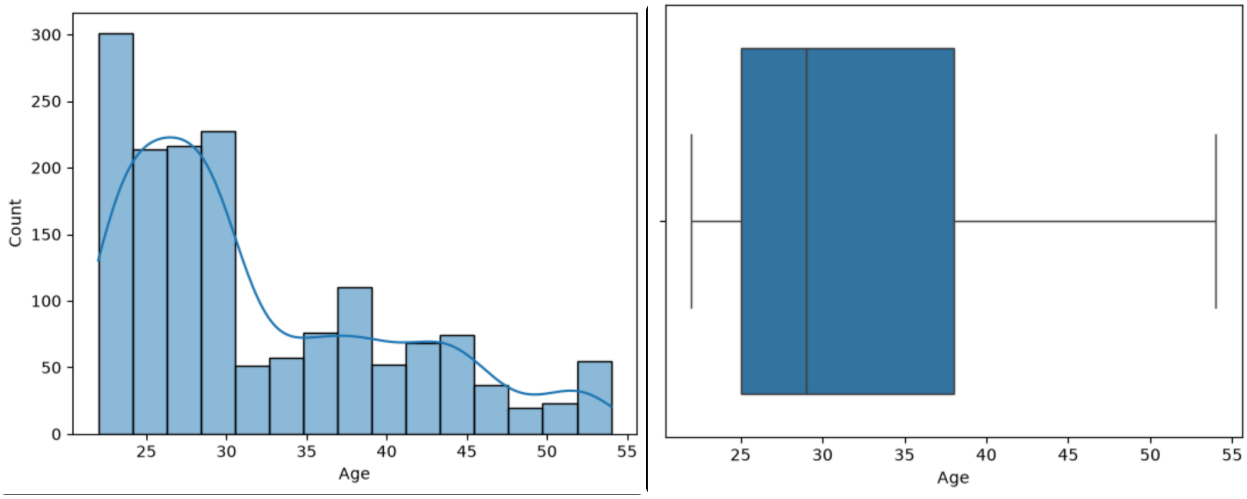


Fig9

Observation:

- we can see a high number of people in mid 20s to 30s indicating customers that are young and potentially first time buyers
- median is around 29, and majority of the customers again here are clearly indicating ages of around mid 20s to late 30s

Gender

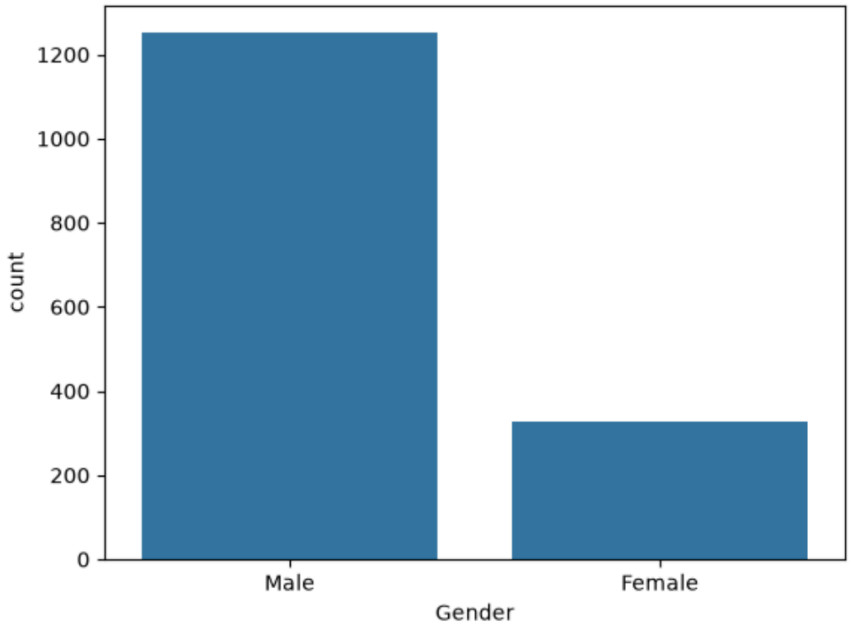


Fig10

Observation: Males seem to be the dominant Gender in the set of Cusotmers

Profession

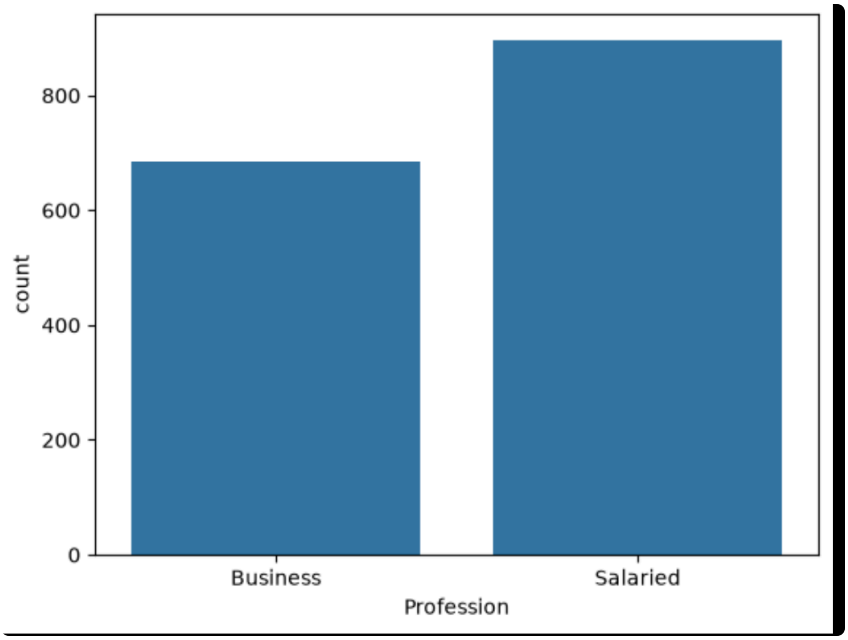


Fig11

Observation: More number of Salaried Professionals as compared to Business

Marital Status

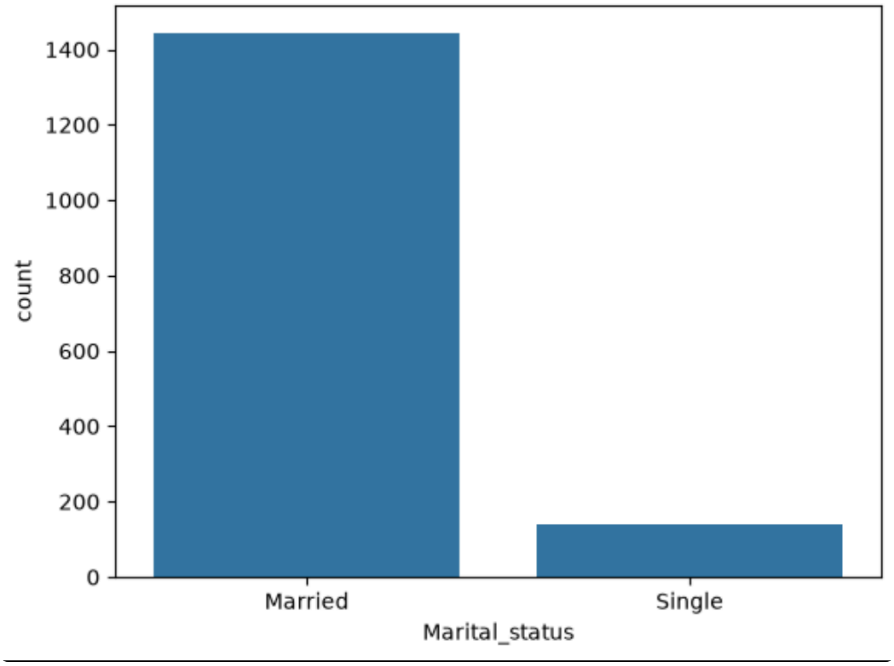


Fig12

Observation: Large amount of ie Majority of Customers are Married

Education

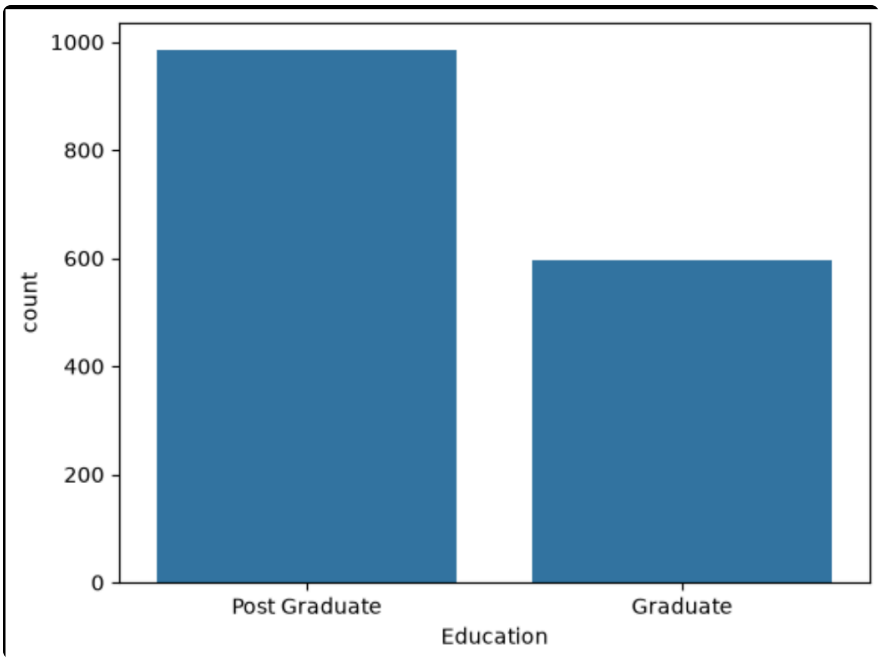


Fig13

Observation: More number of Post Graduates as compared to Graduates

No_of_Dependents

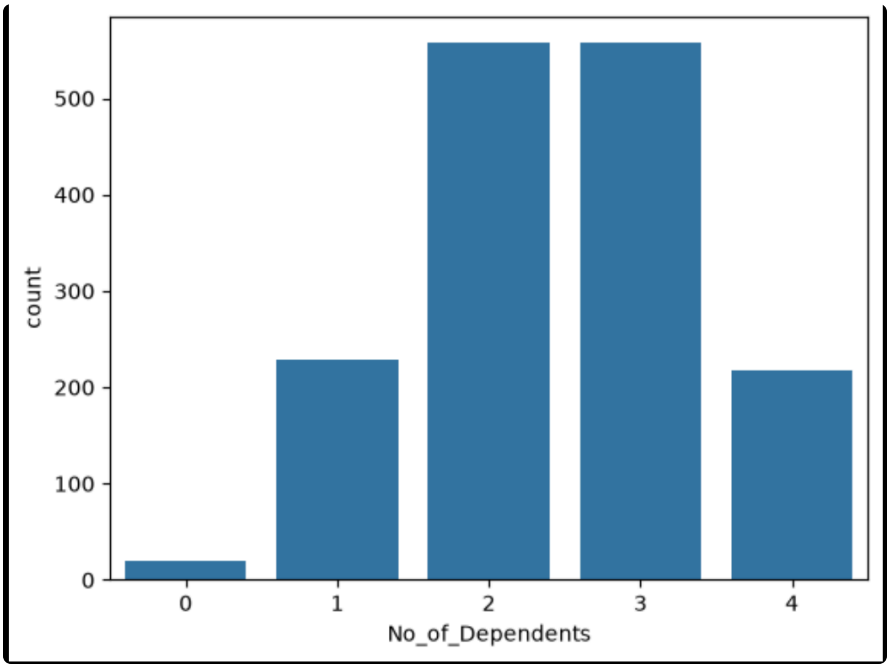


Fig14

Observation: Majority of the customers seem to be having families or about to start a family

Personal_loan

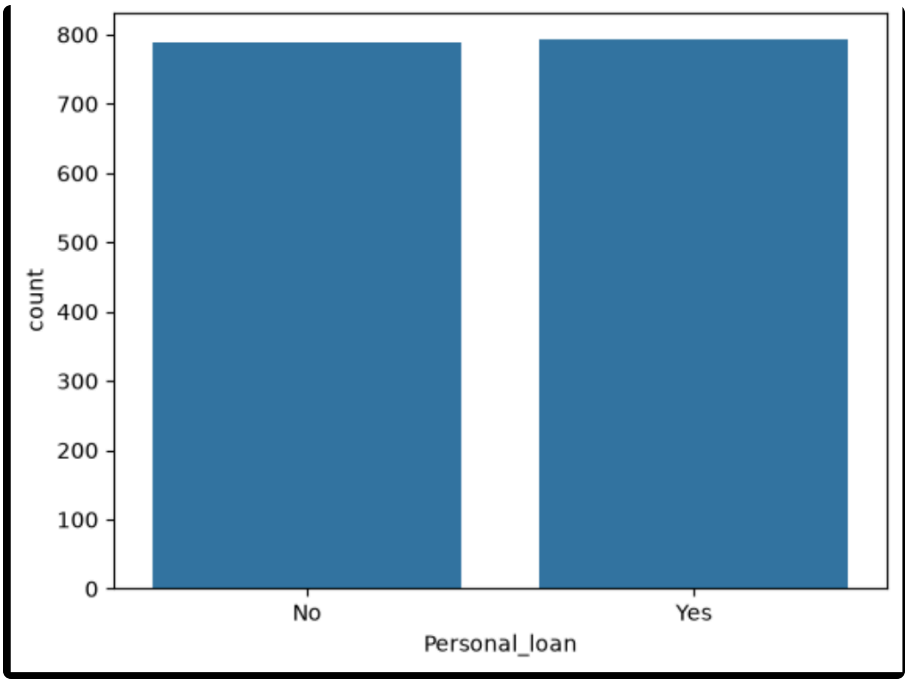


Fig15

Observation: Equal number of people having taken and not taken a personal loan

House_loan

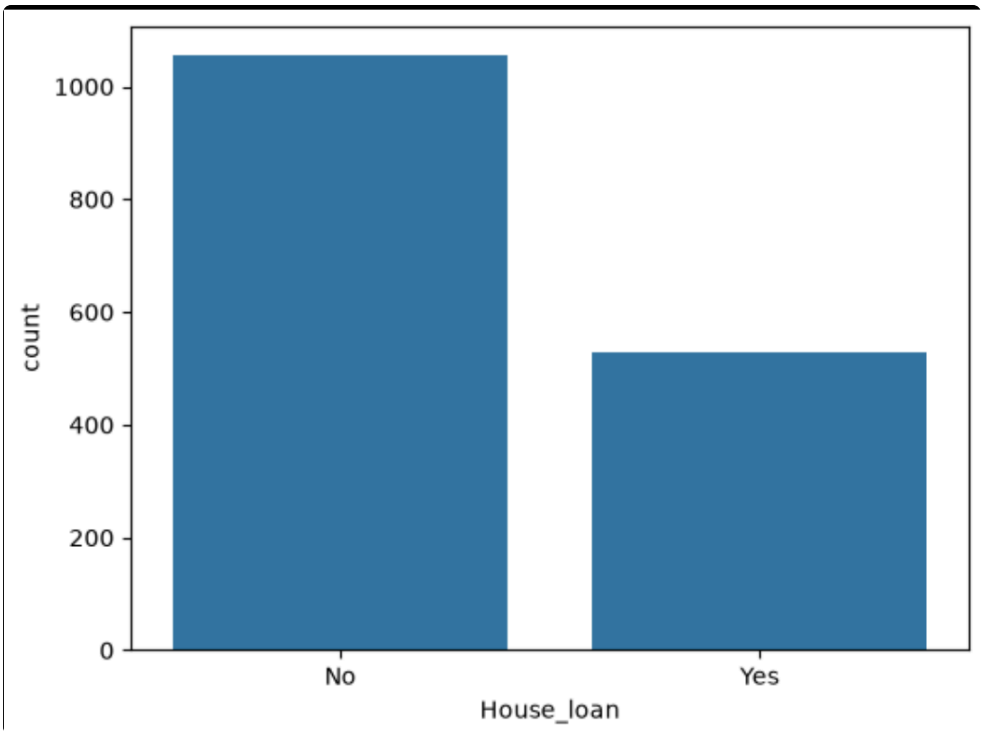


Fig16

Observation: most buyers ie buyers without House loans are in majority

Partner_working

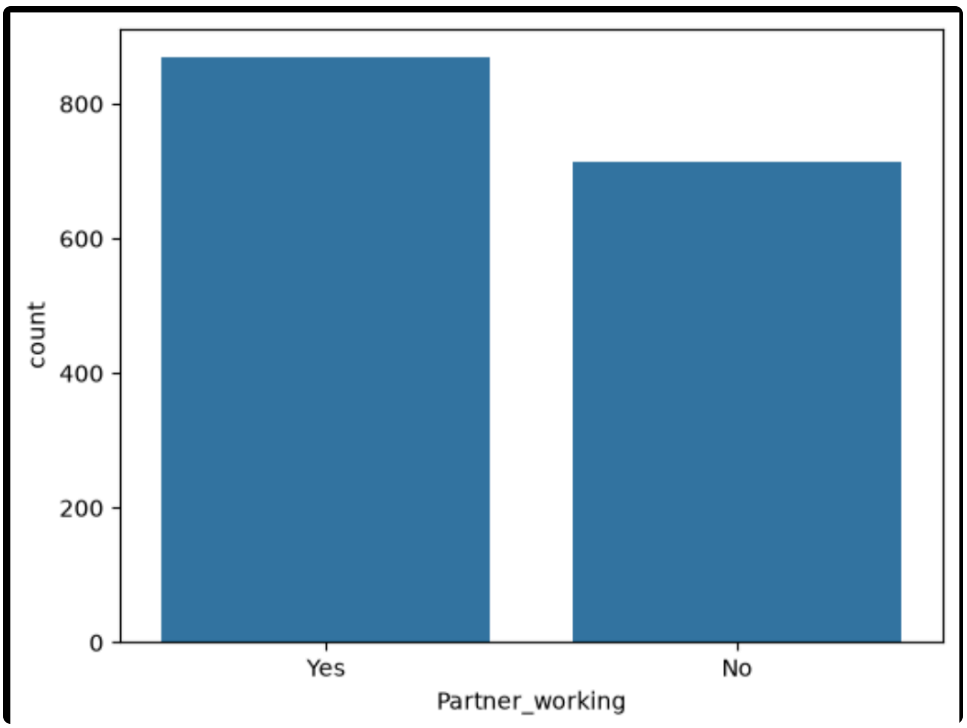


Fig17

Observations: many partners are working , and more than number of Partners not working

Salary

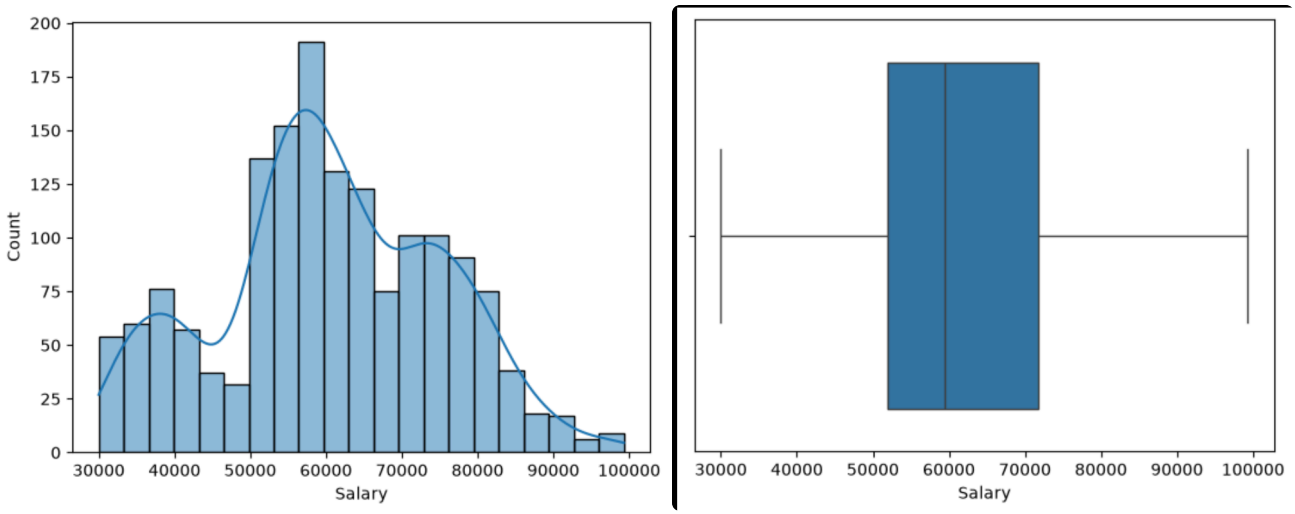


Fig18

Observation: higher concentration of buyers with salaries around 45,000 - 75,000 median being somewhere around 60,000 appealing more to the middle class instead of lower and upper salaries

Partner_salary

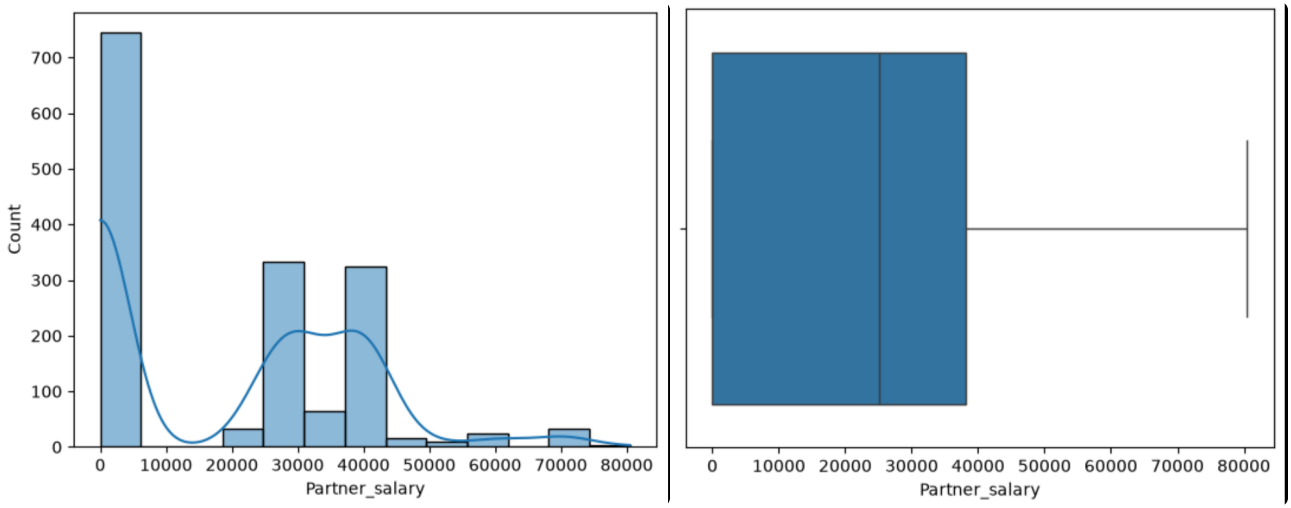


Fig19

Observation: not many partners with a salary or working typical normal range being non to around 40,000 and median near 25,000

Total_salary

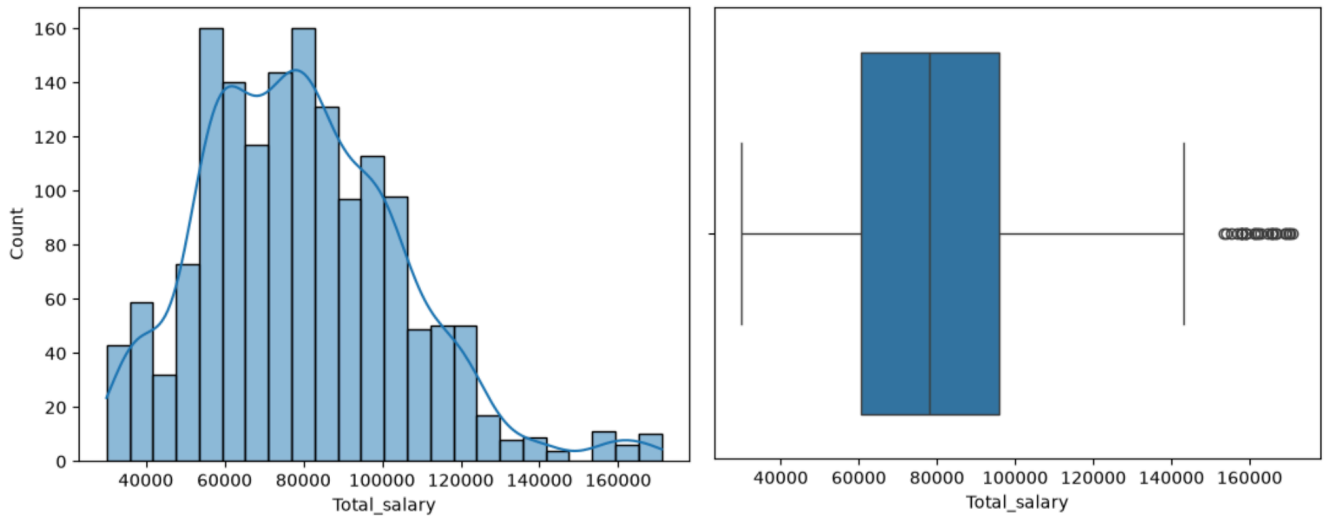


Fig20

Observation: majority of total salary seems to range between 50,000 - 90,000 median being around 75,000 - 80,000 with many outliers beyond 140,000.

Price

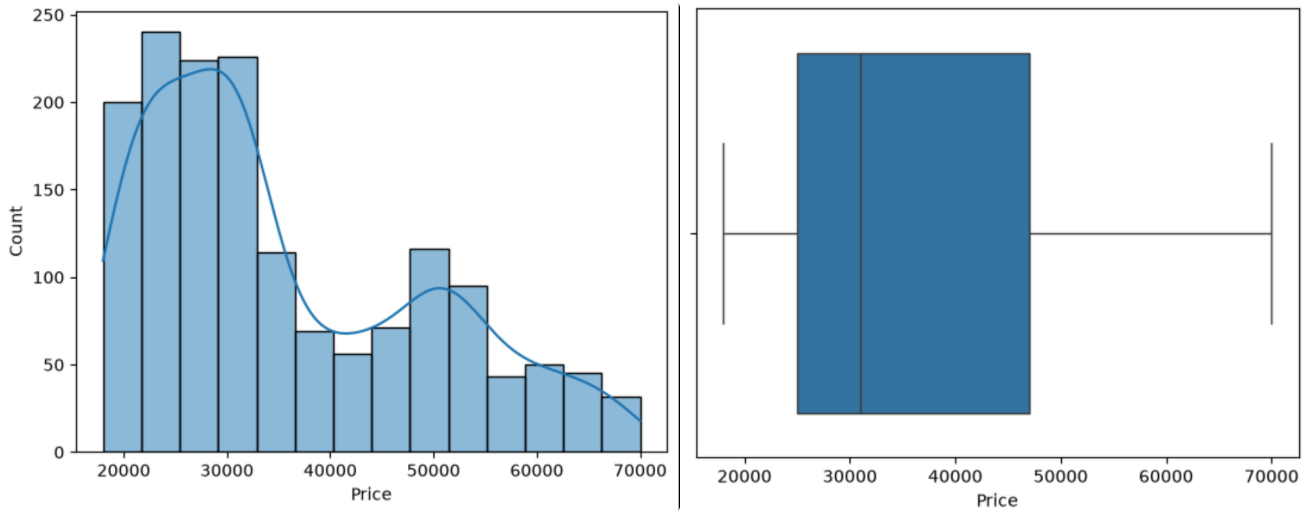


Fig21

Observation: low to mid priced sales seem to be high in number while expensive sales seem to be low due to the low number of high price buyers

Make

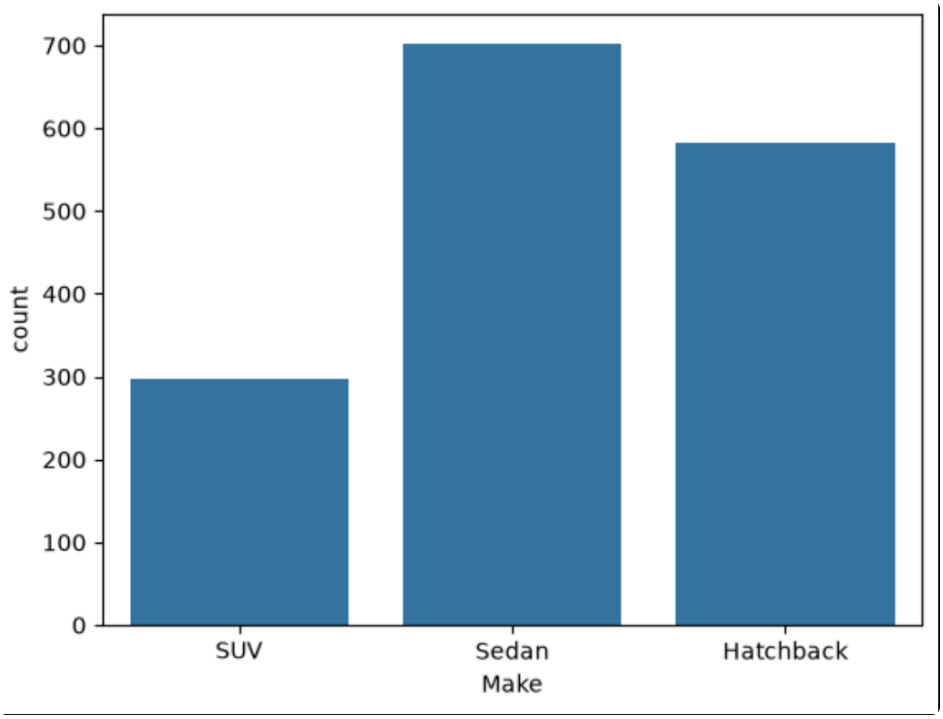


Fig22

Observations:

Observations and insights

- many of the customers are between mid 20s to 30s , indicating to potentially firsts time young buyers
- Majority of Buyers are Male
- Salaried Professional seem to outnumber Business Professionals by some noticeable number of people
- Majority of the customers are married, adding up to the fact that buyers are young, they could be a couple , or are about to start a family
- Post Graduates are in Majority by a descent margin as compared to Graduates
- Given that Majority of the Customers are Married and Young, from the No of dependent visualization we can say that Almost all customers are already having a family or are couples that will start a family, pointing to a customer base looking towards a family oriented car
- Salaries ranging between 45,000 - 75,000 seem to be having high number of sales with median salary being somewhere around 60,000
- There seems to be a large number of customers with no partner salary support but Total Salary Distribution as compared to the Salary Distribution seems to indicate that a large number of sales occur due to Partner Salary support
- There are many Outlier Salaries which seem to be an outcome of low number of High salaried or High Total salary individuals

- Low to mid Priced Caras seem to making a large number of sales, while there are some upper mid sales that are high

3.2Bivariate Analysis

Our Target variables meaning variables that reflect the outcome of marketing campaign and Customer experience outcomes are `Price` and `Make`

Numerical vs numerical

Age vs Price

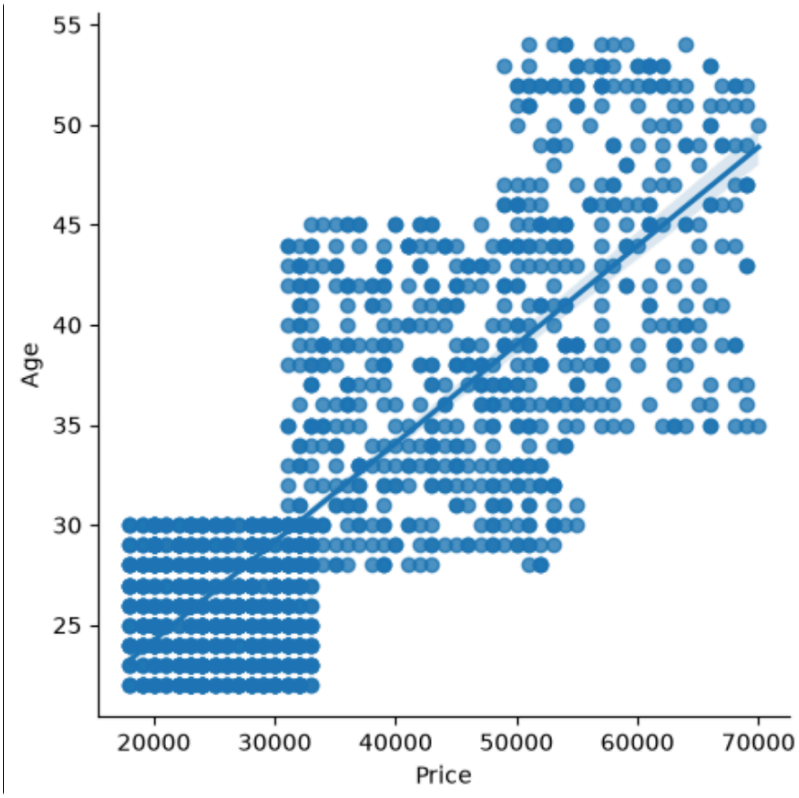


Fig23

Observation:

- Strong Positive Correlation between Age and Price of vehicles
- High concentration of Young Customers Below thirty Purchasing vehicles not more than certain around near 33,000

No_of_Dependencies vs Price

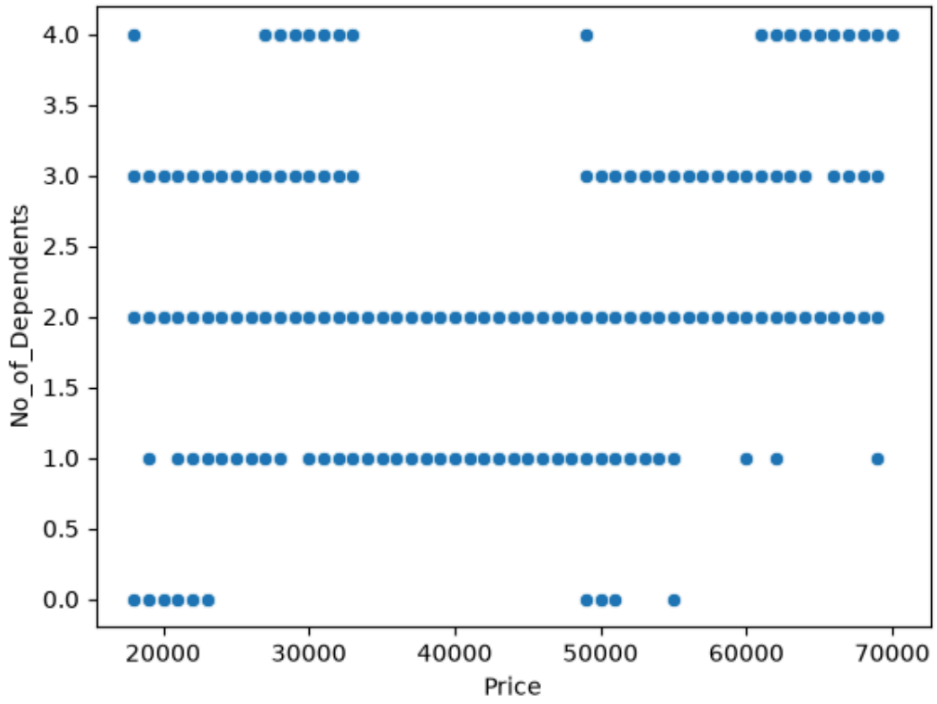


Fig24

Observation: # weak correlation , as majority of buyers have dependents, with individual single solo customers either purchasing lower priced or upper mid priced vehicles ie 20,000 and 50,000

Salary vs Price

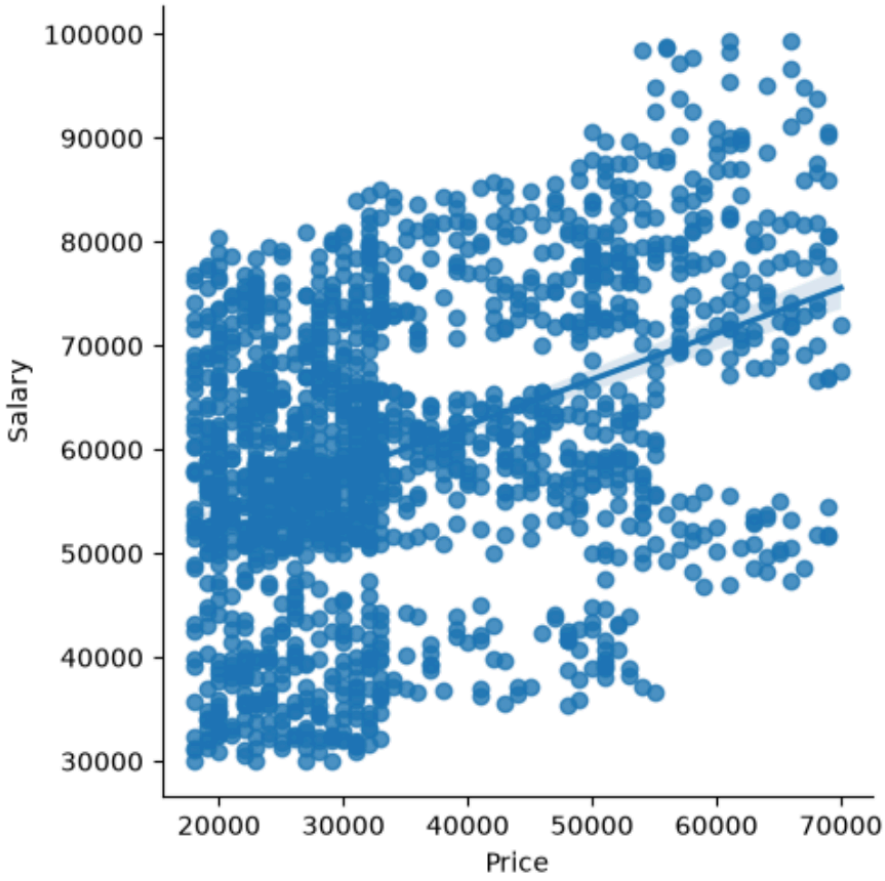


Fig25

Observation: weak correlation but we can see that there is a high concentration of people with low and high salary not going for above around 33,000 , while we can also say that people with moderately low salaries can purchase for upper mid priced vehicles around 50,000 -55,000 , while people with salaries around 50,000 - 80,000 will go for high priced vehicles of 70,000 and people with salaries more than 80,000 -85,000 will purchase high price vehicle around 60,000 -70,000

Partner_salary vs Price

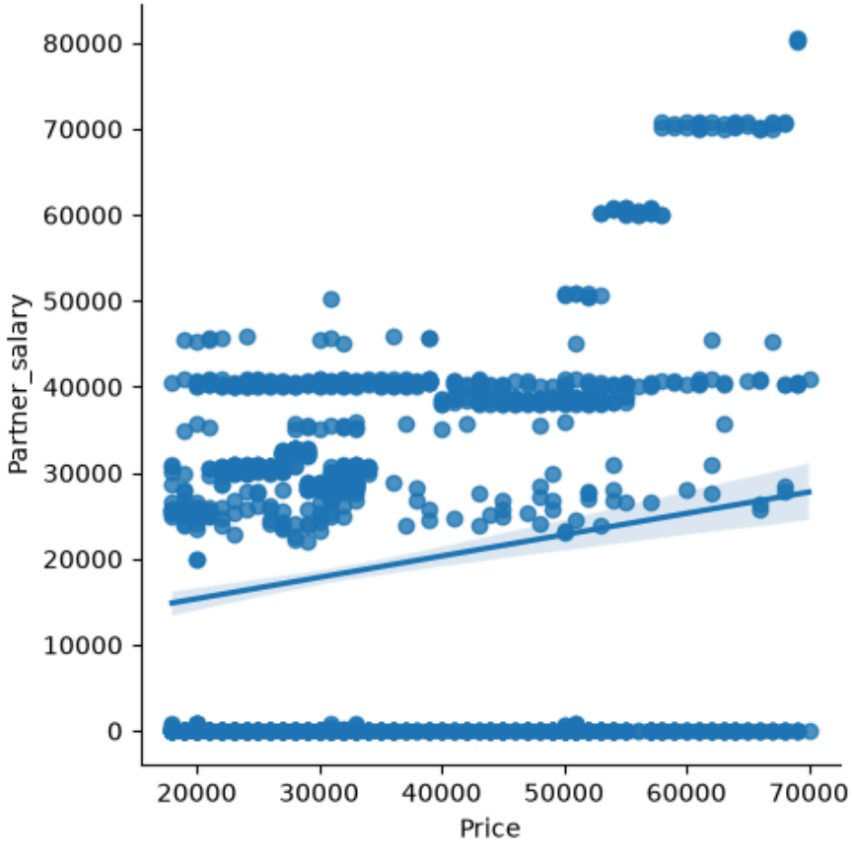


Fig26

Observation: weak correlation , but we can see the across all price ranges of cars there can be and cannot be partner salary but there is a high concentration of buyers with partner support of around 20,000 - 35,000 purchasing a car priced between 20,000-33,000 , but there seems to be a high number of Partner support of 40,000 across all prices of cars

Total_salary vs Price

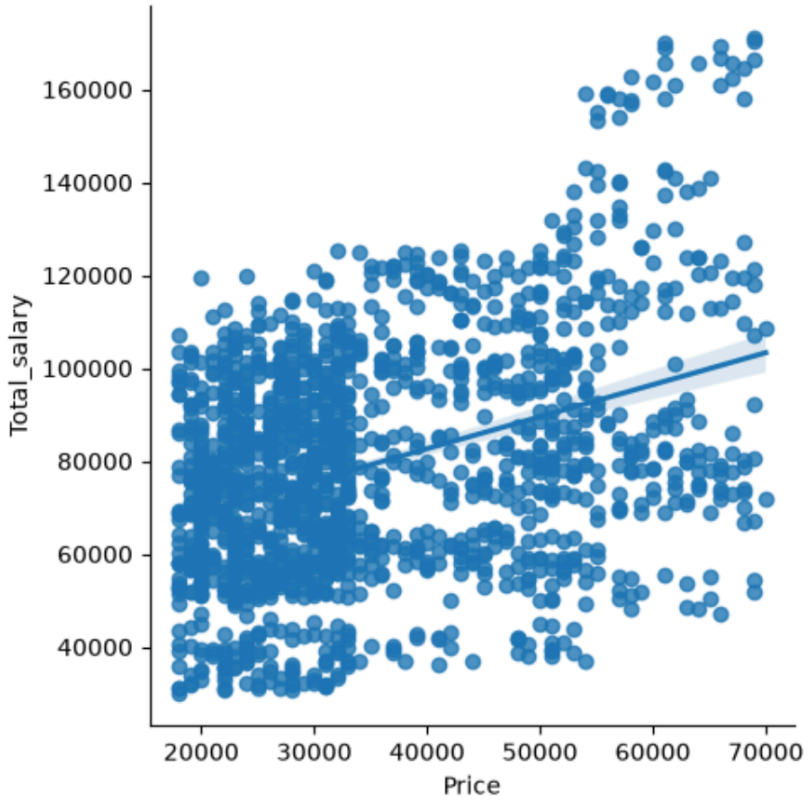


Fig27

Observation: quiet similar to Salary where there is a high concentration of people with total Salary ranging from low to high 100,000 -105,000 not purchasing more than around 33,000, which again seem to point to certain attraction , appeal and value for money on the 33,000 price limit

Observations and Insights overall

- There seems to be strong correlation between price and vehicles, whit high concentration of Young Customers choosing to keep their budge cap to 33,000
- in the case of single customers, they will either purchase a lower priced or upper mid priced vehicle ie 20,000 or around 50,000
- large number of people with salaries ranging from lower to upper mid 80,000 seem to have a budge cap around 33,000
- Across the whole price range for cars there are customers with no partner support, while noticeable amount of customers with partner support of around 20,000 - 35,000 purchasing a car priced between 20,000 - 33,000, but there seems to be a high number of Partner support of 40,000 across all prices of cars
- There is certain appeal to price limits of 33,000 for buyers with salaries ranging till 100,000 -105,000

Correlation

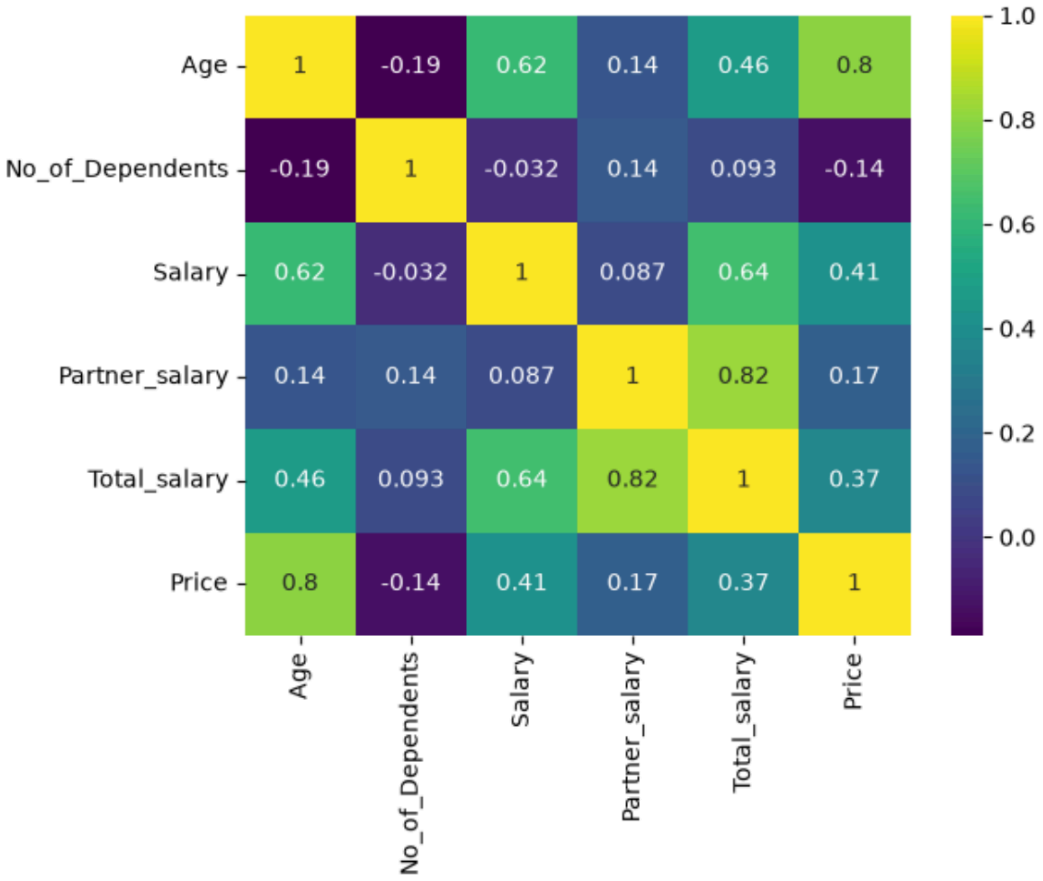


Fig28

Strong Correlation between the following variables

- Total_salary and Partner_salary
- Total_salary and Salary
- Salary and Age
- Price and Age

Noticeable correlation:

- Total_salary and Price
- Total_salary and Age
- Price and Salary

Categorical vs Numerical

Age vs Make

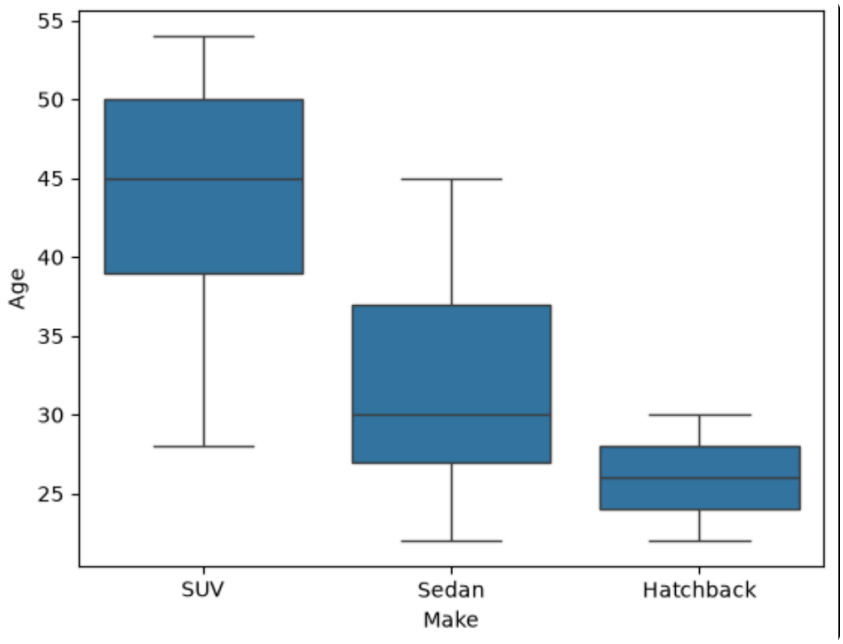


Fig29

Observations: here we can see the upper middle aged customers seem to prefer SUVs, while very young customers prefer hatchbacks , while young to middle aged adults seem to prefer Sedans, and majority Senior Customers seem to lean more towards SUV

No_of_Dependents vs Make

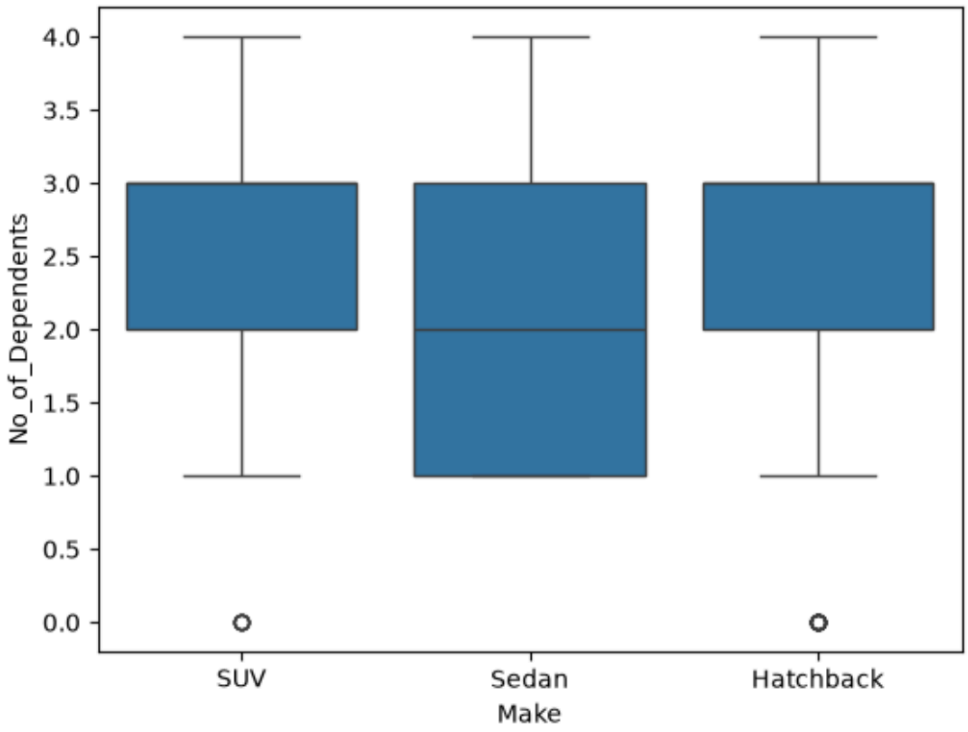


Fig30

Observation:

- all 3 categories seem to typically have dependents being either 2 or 3 , with Sedans having no dependents as 0, indicating that nearly all sedan buyers are not single , and are either a couple that can start a family or have already started a family
- while single customers seem to be choosing SUVs or Hatchbacks indicating that sedans seem to be attracting customers wanting a car for the family

Salary vs Make

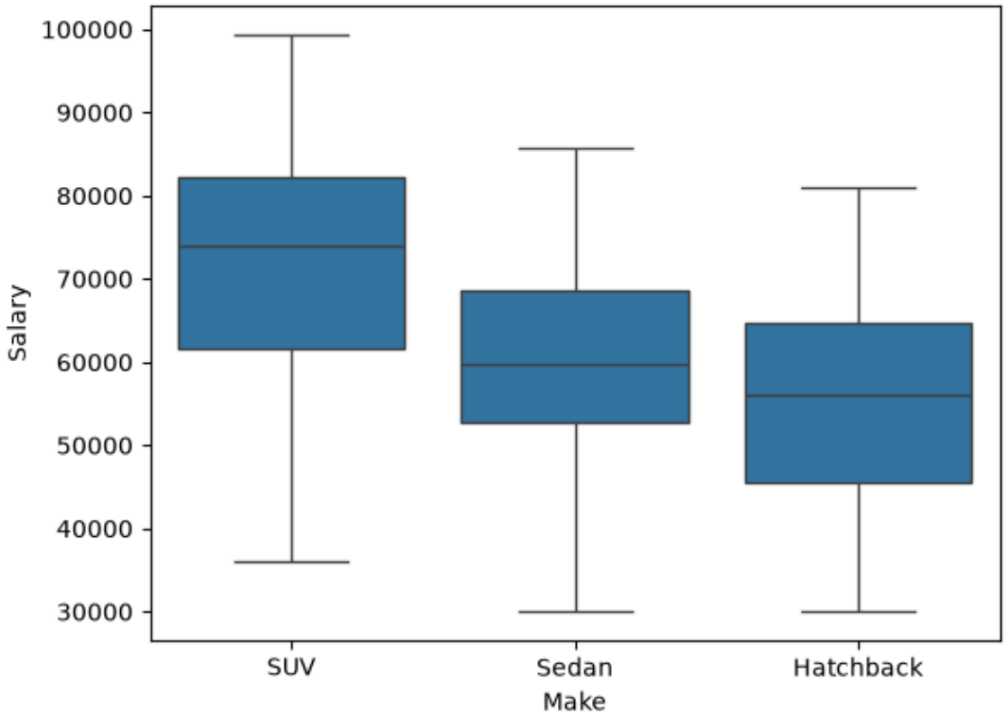


Fig31

Observations:

- a similar trend of like Age is observed, where people with more salary seem to be taking SUVs while sedans seem to be appealing to mid salary earners, and low salary earners choosing hatchback
- this seem to reflect the correlation between Salary and Age ie as someone Ages their Salary increases , so from here we can say that senior adults Tend to have a high salary and tend to choose SUVs

Partner_salary vs Make

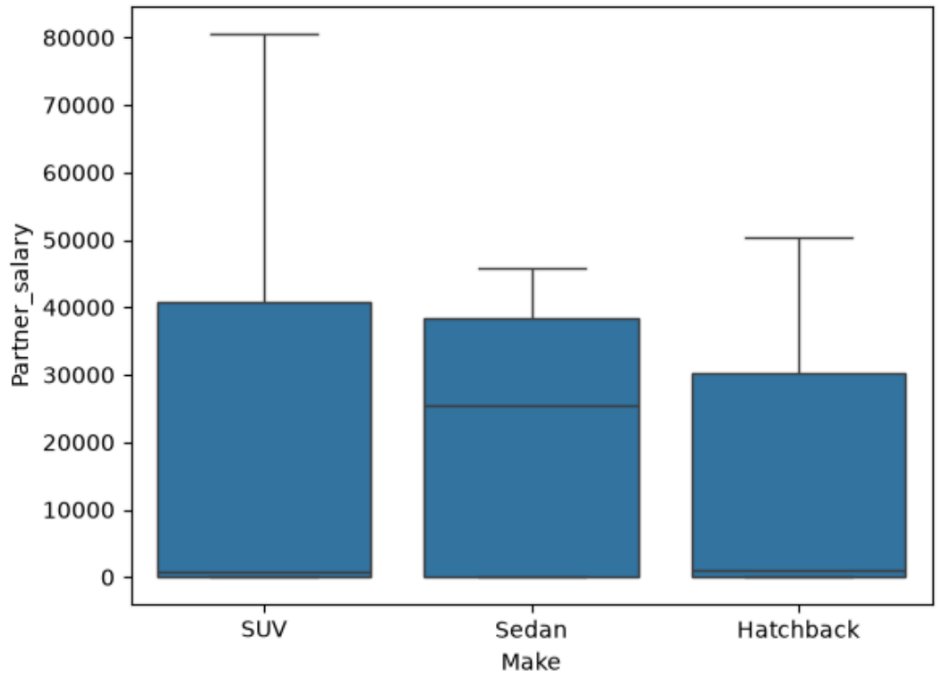


Fig32

Observation: in the case when Partner Salary is high , the customer will tend to go for SUV, people with Partner Salary seem to leaning more towards Sedan ie 50% of customer with median salary of 25,000 will go for Sedan while SUV and hatch back have 50% of sales with no partner support

Total_salary vs Make

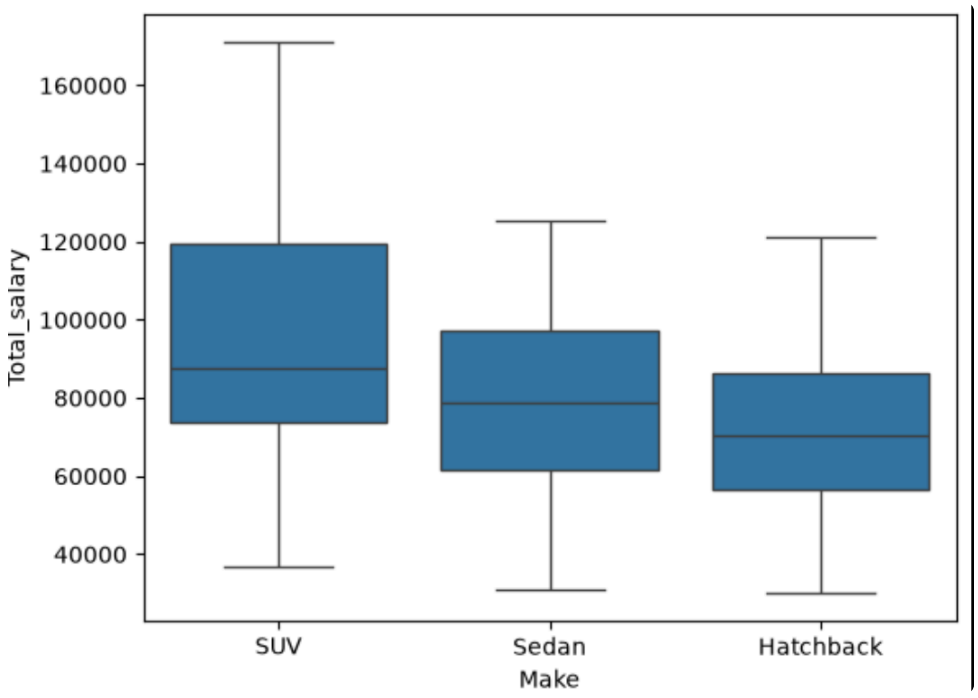


Fig33

Observation: a slightly similar trend as observed in the visualization with Salary and Age, but median Total Salary seem to be pretty close but still in descending order, but clearly people with very high total salary will go for SUV, indicating some form of premiums, quality , luxury, and size, for High earning Senior Customers

Price vs Make

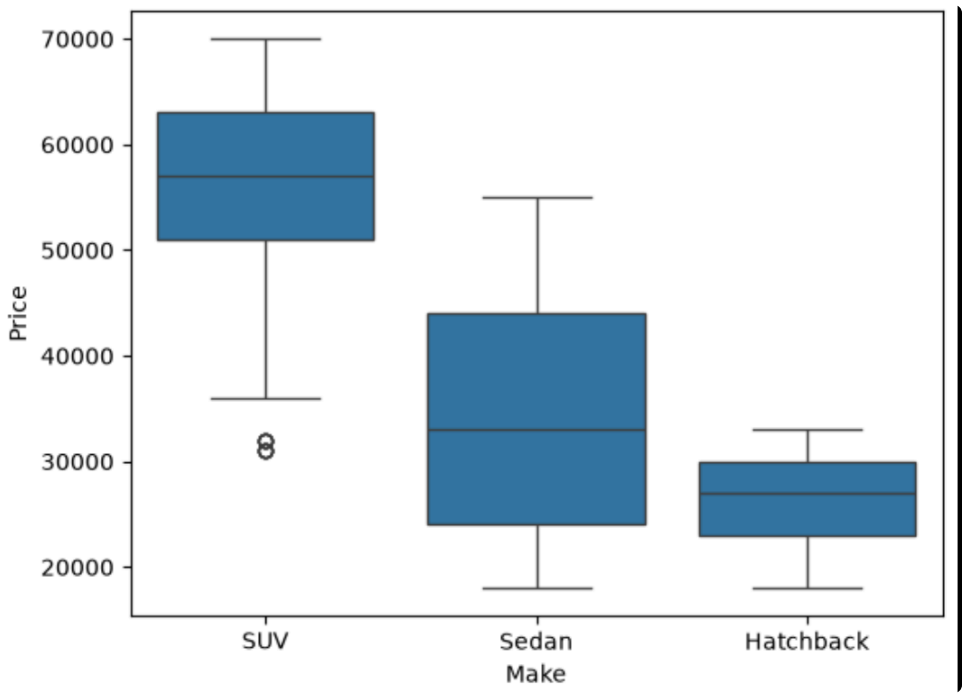


Fig34

Observation:

- as shown SUVs are more expensive as compared to Sedans and Hatchbacks, with prices of SUV starting from mid pricing to High Pricing, but majority of SUV buyers purchasing around the 50,000 - 60,000 bracket, while the starting price of hatchbacks and SUVS are nearly the same, with Sedan Prices having the high priced options
- this would indicate that SUV buyers, mostly Senior Adults with Upper mid to High Salary , and already have family members that are teenagers or above 18, and possibly require large amount of space for accommodating members of the family that are mostly likely no longer smaller sized occupants

4 Key Questions

1. Do men tend to prefer SUVs more compared to women?

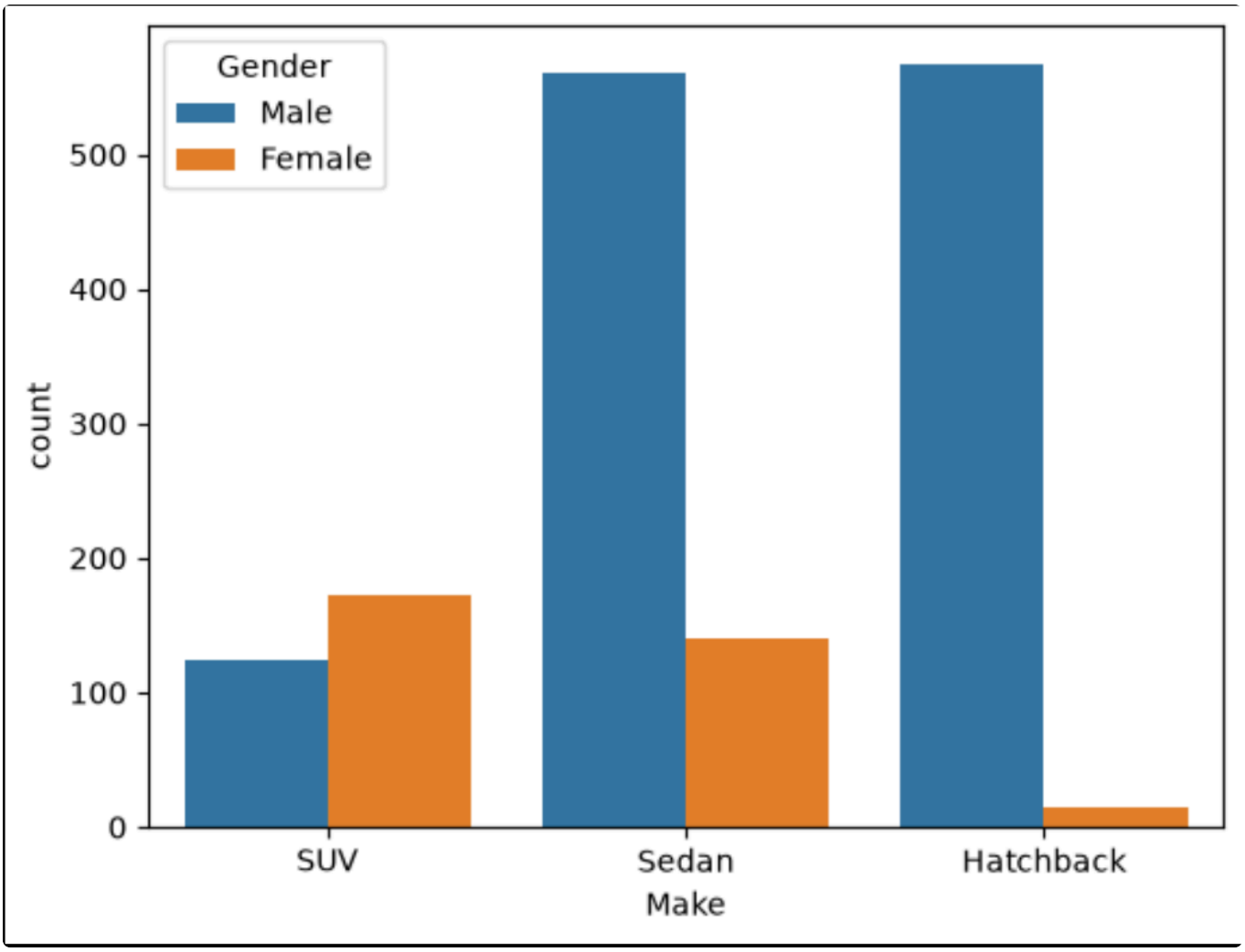


Fig35

No women tend to prefer SUV over men

2. What is the likelihood of a salaried person buying a Sedan?

Given Below is a plot of percentage of people specifically Salaried People who have purcsed each category of vehicle

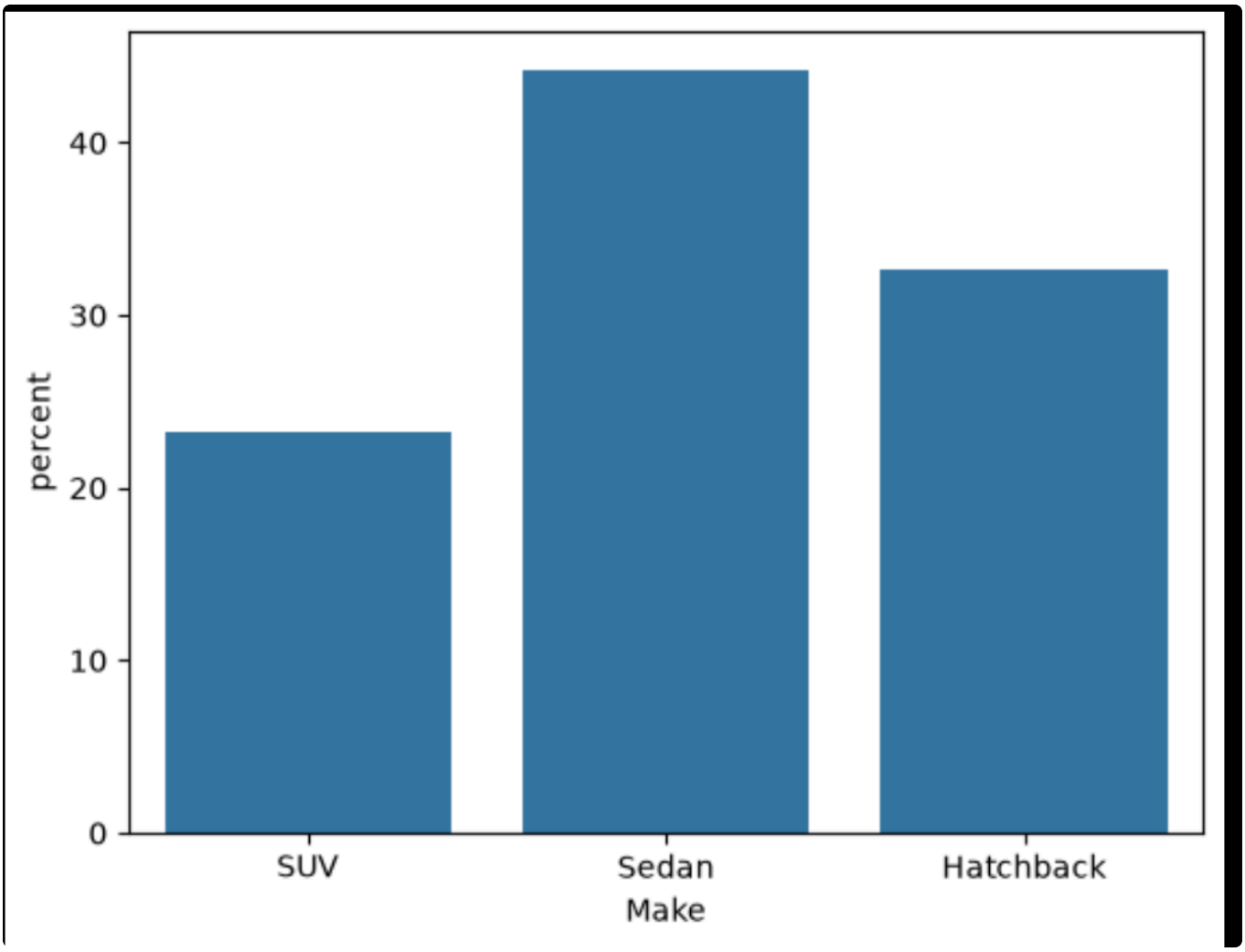


Fig36

Around 45 percent its likely that a salaried person will choose a sedan

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

The same plot as the previous one but this one is specifically for Males

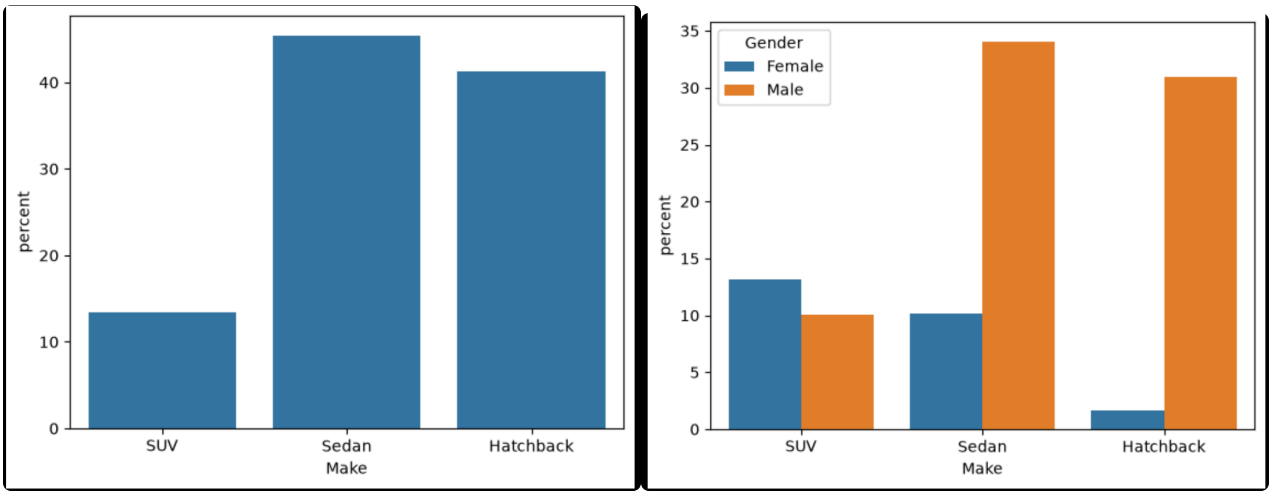


Fig37

no evidence support Sheldon Coopers claim exists, instead counter evidence of Males most likely choosing Sedan or Hatchback exists and Salaried Males are not an easier Target for SUV Sale Over a Sedan Sale

4. How does the the amount spent on purchasing automobiles vary by gender?

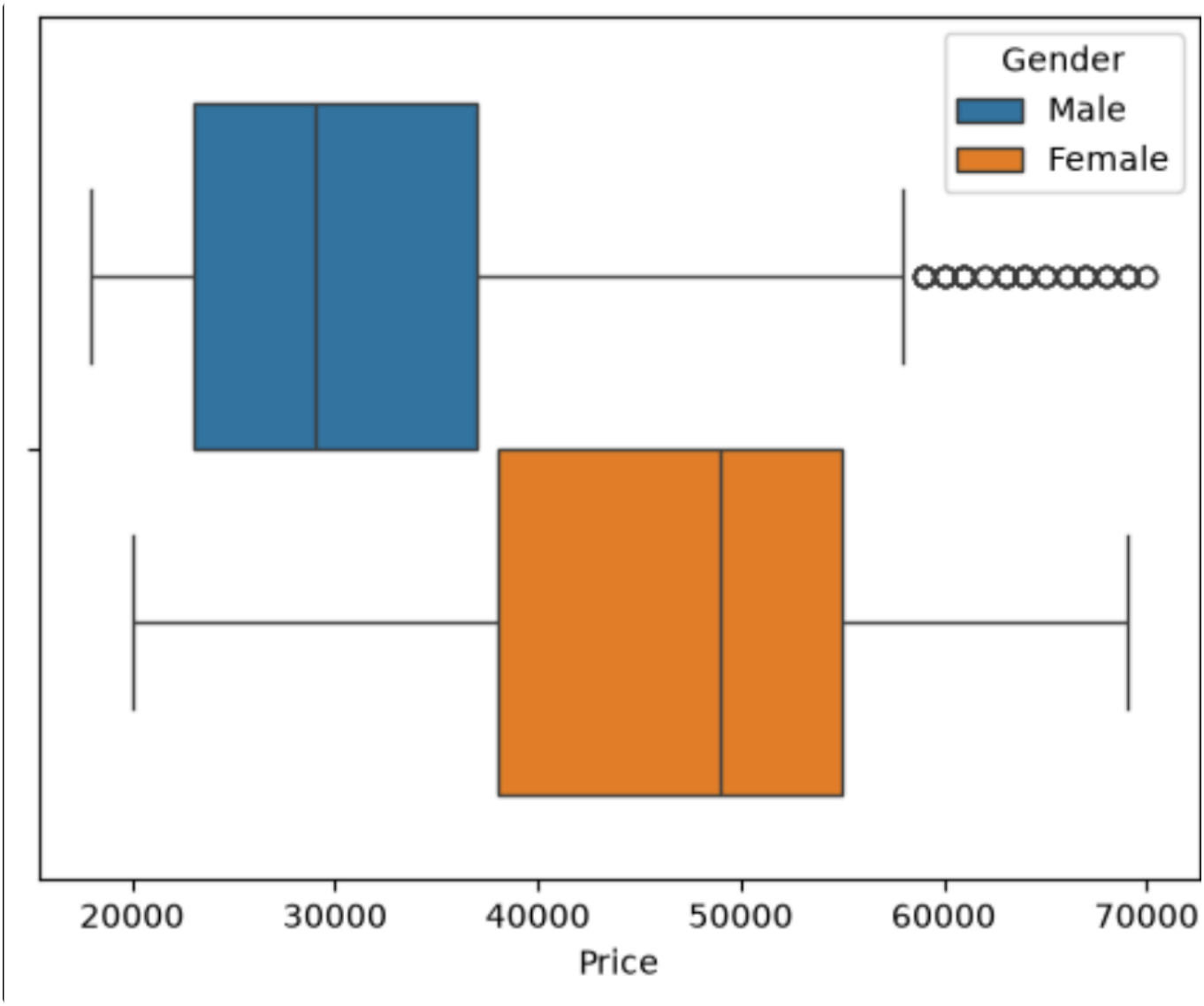


Fig38

- 50% of females tend to purchase cars Priced around close to 50,000 while in the case of Males 50% of them tend to just around 30,000
- Its Normal for Females to purchase low to high priced vehicles, while its uncommon and rare for males to spend about more than 57,000

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Total money that was spent on purchasing automobiles by individuals who took a personal loan is 27,290,000

6. How does having a working partner influence the purchase of higher-priced cars?

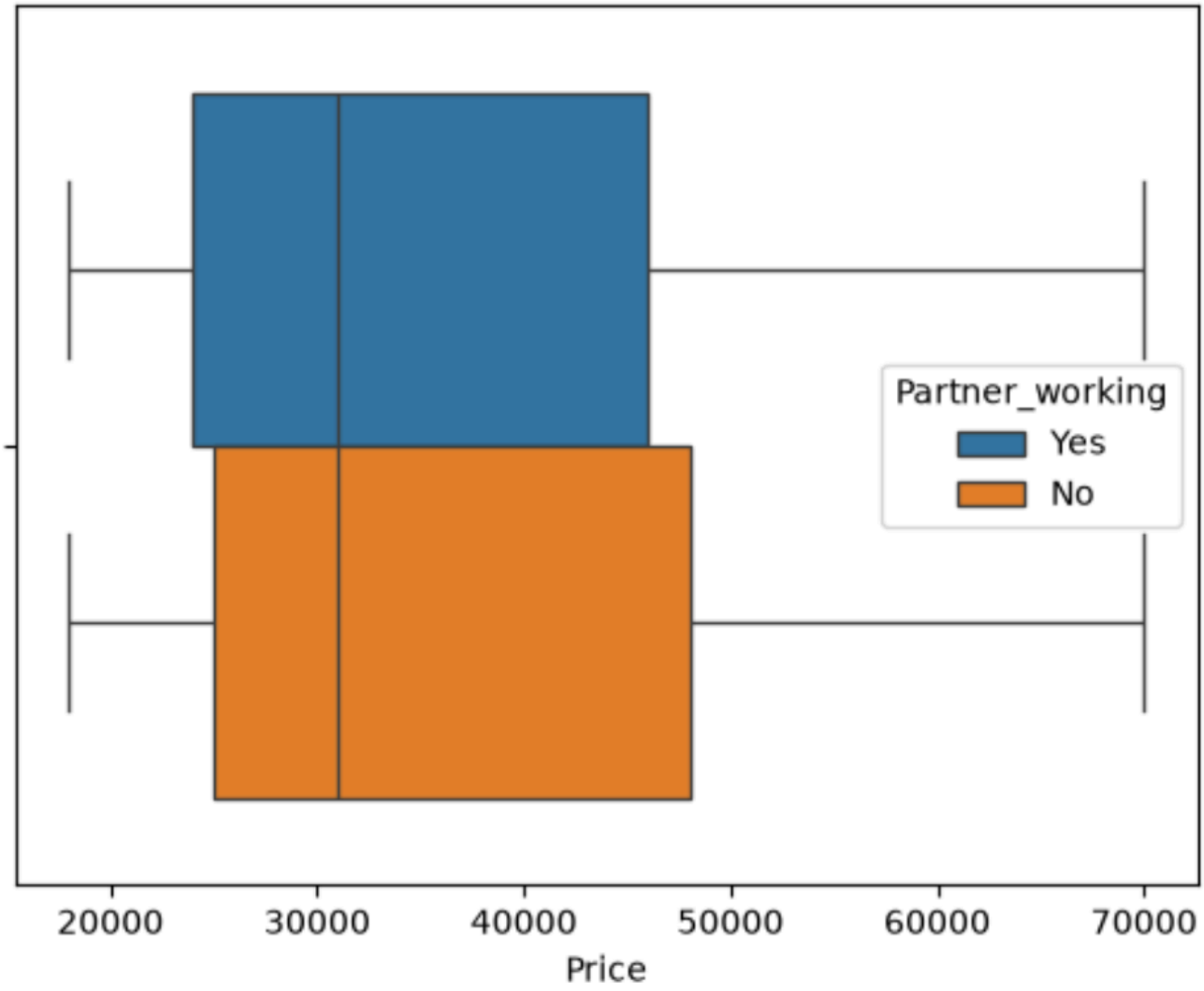


Fig39

Having a working partner does not in any meaningful way affect the purchase of higher cars therefore zero impact

5 Actionable Insights and Recommendations

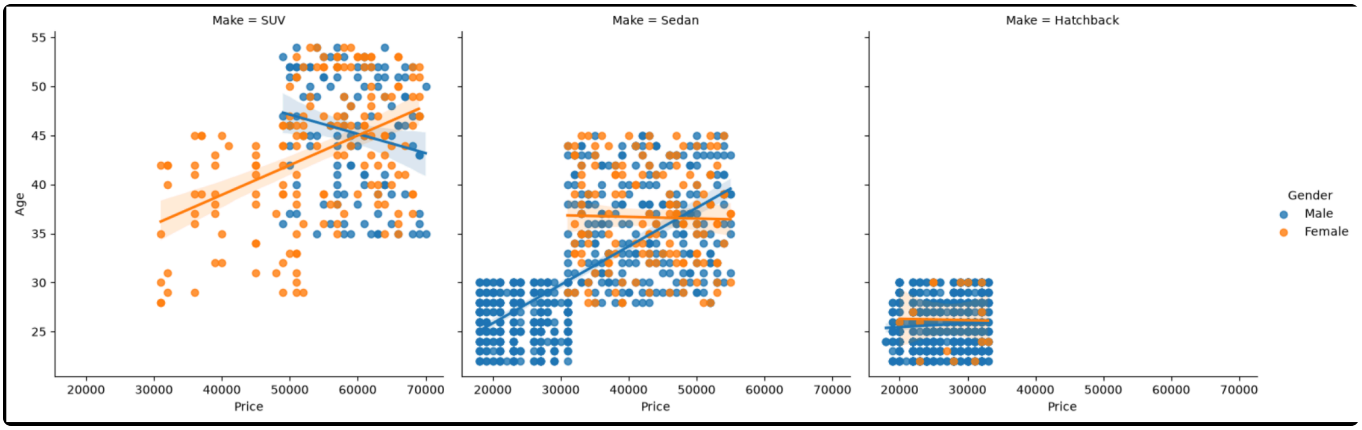


Fig40

- Its observed that Majority of the customers are sedan and hatchback buyers, and from our analysis with the No_of_Dependents, all categories of Make have customers with family

  Therefore a car can be manufactured with a focus on attractive features and gimmicks that other family members will enjoy, especially young customers specifically for hatchbacks and Sedans
- Majority of buyers ie a large number of them are young, and by our overall assessment of No of Dependents, we see that they are single, or married and can possibly start a family indicating them being first time buyers

  Hence a Focus on family friendliness, and appeal should be made on the Manufacturing of the vehicles, along with durability and reliability, likable features.
- From Fig 41 as shown above SUVs seem to have lower sales with Female Customers being more than Male, and Prices not being at the low end for people to find it appealing enough

  Pricing of SUVs can be set around the same Price Range for Sedans, so that it appeals New young Buyers that have a typical low budget if given the option for SUVs that are priced in a similar Range like Sedans should help boot SUVs Number of Sales
- Women seem to be more appeased with SUVs and Cars

  So marketing can focus on features and characteristics of cars that appeal to women, such as safety features, vehicle tracking, easy to drive, fun, comfortable, and confidence inspiring so that sales for SUVs can increase especially due to women
- For Hatchbacks and Sedan Majority of the Customers seem to be Men that are Young and are Married or about to start a family .

  Hence a focus on Family oriented, fun to drive, strong performance , reliable, comfortable.
- Sedan Buyers seem to have more support from Partners, and hence highest number of sales

  Extra incentives to promote buying of other category of vehicles can be made to boost sales especially SUVs
- Introduction of Loyalty Programs to leverage this high volume of young first time buyers, so that at some point in the future they end up upgrading to a new vehicle from Austo Motor Company only such that they end up purchasing a new upgraded Category, therefore supporting growth of sales in SUVs and Sedans